Cardiac MRI Segmentation With a Dilated CNN Incorporating Domain-Specific Constraints

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***Abstract*—Semantic segmentation of cardiac MR images is a challenging task due to its importance in medical assessment of heart diseases. Having a detailed localization of specific regions of interest such as Right and Left Ventricular Cavities and Myocardium, doctors can infer important information about the presence of cardiovascular diseases, which are today a major cause of death globally. This paper addresses the problem of semantic segmentation in cardiac MR images using a dilated Convolutional Neural Network. Opting for dilated convolutions allowed us to work in full resolution throughout the network's layers, preserving localization accuracy, while maintaining a relatively small number of trainable parameters. To assist the network's training process we designed a custom loss function. Furthermore, we developed new augmentation techniques and also adapted existing ones, to cope for the lack of sufficient training images. Consequently, the training set increases not only by amount, but by substance as well, and the network trains quickly and efficiently without overfitting. Our pre- and post-processing steps are also crucial to the whole process. We apply our methodology for the Right and Left Ventricles (RV, LV) and also the Myocardium (MYO) according to the Automated Cardiac Diagnosis Challenge (ACDC) with promising results. Submitting our algorithm's predictions to the Post-2017-MICCAI-challenge testing phase, we achieved similar scores (average Dice coefficient 0.916) on the test data set compared to the state of the art featured in the ACDC leaderboard, but with significantly fewer parameters than the leading method. Our approach outperforms other methods featuring dilated convolutions in this challenge up until now.**

***Index Terms*—Cardiac MRI segmentation, CNN, data augmentation, loss function, dilated convolutions, MRF.**

I. Introduction

AUTOMATED semantic segmentation in medical images is a very challenging task. When performed correctly, it can assist doctors who would be spending a significant amount of time to manually segment regions of interest (ROIs). In addition to saving a lot of time, it eliminates ambiguities from human intervention, which can occur when the regions are annotated by more than one experts. Furthermore, it can serve as a first step towards automatic disease diagnosis. In particular the human heart can suffer from various cardiovascular diseases, still a major, if not the most significant, cause of death globally.

In recent years, Deep Convolutional Neural Networks (CNNs) have achieved state-of-the-art results in biomedical image segmentation tasks. Especially the U-net architecture [15] proved to be task independent and has been widely applied with minor or major modifications. Approaches with CNNs for the segmentation of the Right and Left Ventricles, as well as the myocardium outperformed other more conventional computer vision and machine learning approaches [1]. The majority of the best performing algorithms for ventricular segmentation are inspired by the U-net architecture.

Dilated, or *a trous*, convolutions have also been proven quite effective in capturing context relations in images by drastically enlarging the CNN's receptive field [4]. They have been used as front-end modules [19], as well as in conjunction with residuals [20]. In [18] a dilated CNN in a relatively shallow architecture is used standalone in the framework of the Automatic Cardiac Diagnosis Challenge. In [9] the U-net is given a context extractor built using a dense *a trous* convolution block and the new network outperforms the U-net in many medical applications.

The dilated CNN with 3×3 convolutions is known to suffer from gridding artifacts [20]. In the residual dilated networks the gridding artifacts have been alleviated by adding more layers after the block of dilated convolutions. In [17] two methods of smoothed convolutions are proposed for addressing the gridding artifacts.

In this work, we address the task of semantic segmentation of cardiac MR images having as main objective and motivation to obtain a network that is simple as possible, driven by the application domain. We propose a simple architecture with cascaded dilated convolutions. In semantic segmentation, and moreover in cardiac segmentation, localization performance is primordial. We have opted for a dilated CNN in order to preserve image resolution. In this framework we propose to use large kernel size filters at the early layers for obtaining learnable smooth convolutions. As in our design the kernels are larger at the first layers and become narrow at the last layers, dense connections between the feature maps are established through the layers.

The deep neural networks rely heavily on big data to have good generalisation performance. Unfortunately, our targeted application domain does not have access to big data, as the data annotation is very costly and often requires significant amounts of expertise. A data-space solution to the problem of limited annotated data is data augmentation. Data augmentation

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encompasses a suite of techniques that enhance the size and quality of training data sets, such that better deep learning models can be built using them [16].

We further establish, as also shown in [18], that dilated convolutions do not only serve as front end modules or intermediate blocks to other networks, but are also capable to perform an advanced task such as semantic segmentation in a standalone architecture. The main contributions of our work are summarized as follows:

1) We propose a simple and efficient filter-bank size and filter kernel size design for smoothing the *a trous* convolutions and minimizing the gridding artifacts, while strengthening the expressiveness and the interpretability of the early network layers.
2) We propose a new data augmentation method adapted to the cardiac MRI data.
3) We enrich the loss function with criteria taking into consideration anatomical constraints for heart structure.
4) We also propose domain specific post-processing modules dealing with unavoidable anatomical errors in deep learning segmentation.

Having determined a region of interest where both ventricles appear in a specially designed pre-processing step presented in subsection II-A, we train the network end-to-end using specific data augmentation procedures presented in II-B, to cope with the lack of training data and boost training efficiency. The detailed presentation of the CNN architecture with the rationale of the filter-bank design is presented in subsection II-C, while the new terms of the loss function are given in subsection II-D. We use data adapted augmentation, without neglecting the already existing and successfully widely applied image transformation techniques. Even so, we still keep the number of training images, as well as the training time, in a small scale compared to other approaches. The post-processing modules are presented in Section III. We chose to work on the Automated Cardiac Diagnosis Challenge (ACDC) data set[1] which provides a set of annotated images covering various pathologies. Our experiments and results on the training, as well as the test, data sets are discussed in Section IV.

II. Proposed Learning Method

A. Data Pre-Processing

First the heart is localized using a method presented in [8]. The initial localization is based on the intensity change between corresponding images in the end diastole (ED) and end systole (ES) phases. Then, a slice in mid-cavity and in the ED phase is segmented using the Chan Vese active contours algorithm [3] to localize the two ventricular cavities on the selected slice and obtaining for the whole data set images of size 120×120, reducing the computational cost and improving the accuracy of the results.

Then, the image intensity is transformed by a function resulting from the histogram equalization of the whole Training data set. The transformation (solid line) is given in Fig. 1 in comparison with the function operating a histogram matching with a Gaussian (dashed line) for the specific image illustrated at the left in this figure, while at the right the transformed image is given.

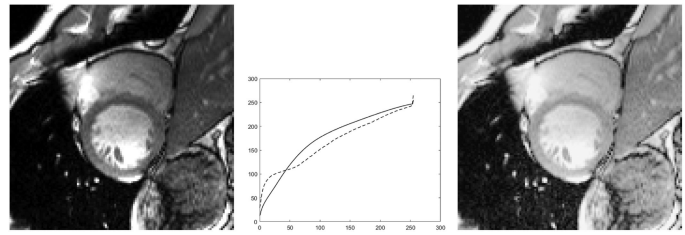

Fig. 1. Image intensity transformation.

B. Data Augmentation

Data augmentation is a technique primarily used when too few training samples are available to train the network. It becomes even more crucial when dealing with medical images, especially when semantic segmentation is involved. Annotating correctly the regions of interest within an image requires specialized expert knowledge and is a heavy time-consuming task. And still, ambiguities arise even when the ROIs are annotated twice by the same expert. This results in having even fewer images to work with, in opposite to other research areas. Having little training data, the network cannot learn the desired invariance and robustness properties or it overfits quickly. To increase the number of training samples, basic image transformation techniques are quite commonly used, like random shifts, rotations, scales, elastic deformations and flips. When applied on the original training data they essentially provide multiple views of the same image.

We design a novel augmentation technique specifically for cardiac MR images based on intensity transformation. In order for the network to learn how to accurately distinguish the left ventricle (LV) cavity from the myocardium, the intensities of these two areas are transformed producing augmented images. These two areas of interest intertwine heavily with each other and, in some cases, are hard to distinguish even with the human eye. At first, the probability density functions of the two areas, myocardium and LV cavity, are estimated. Then, the intensity contrast is reduced adjusted by the Bhattacharyya distance [2] between the two distributions. By also adding noise, more data variation is obtained.

More precisely, if $d_B(p_{Myo}, p_{LV})$ is the distance between the two distributions, the intensity for the pixels in the LV cavity is randomly transformed as

$$I_{LV,aug} = (1 - d_B(p_{Myo}, p_{LV})V + W)I_{LV},$$

while the intensity for the myocardium is transformed as

$$I_{Myo,aug} = (1 + d_B(p_{Myo}, p_{LV})V + W)I_{Myo}.$$

V is a random variable distributed around 0.1 with small variance, and W is a zero-mean random variable uniformly distributed with maximum value 0.05. The additive noise W is pixel-wise determined. Globally the contrast between the

[1] [Online]. Available: https://acdc.creatis.insa-lyon.fr/

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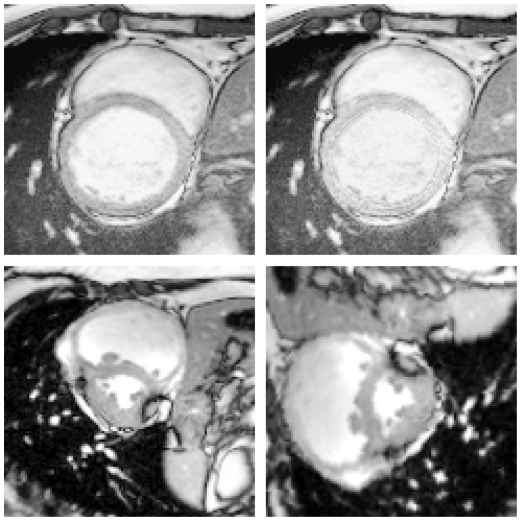

Fig. 2. Original (left) and augmented images (right). Intensities augmentation (top), rotation and zoom in (bottom).

intensity in the LV cavity and the myocardium is reduced, forcing the training algorithm to learn difficult classifications.

To assist our augmentation technique, while also feeding the network with new images never seen before, rotation and scaling in real time is also employed. The rotations are random, but limited to $[0, \pi]$, as we found that the given training set does not feature cases where the LV cavity area and the Right Ventricle are aligned differently. For the scaling process we choose to only zoom in on images that are in the end systole (ES) phase, by a random factor no greater than 1.3. This generates more samples for the end diastole (ED) case. The opposite process of zooming out on images in the ED phase to simulate images in the ES phase, did not help the training process and was therefore finally omitted. At last, whenever ROIs required to perform the previous rotation and scale processes were missing, the real-time augmentation consisted of just a random rotation in the range $[\frac{\pi}{4}, \frac{3\pi}{4}]$.

Original and augmented images are presented in Fig. 2. An example of intensity augmentation is shown on the top and a rotated and scaled sample on the bottom.

C. DCNN Architecture

We propose a simple network of cascaded modules of dilated convolutions, without max-pooling nor any other methods that downscale the feature maps. The convolutions are always performed at full resolution with the network's receptive field covering finally a little bit more than half the size of the input image due to the selected increased dilation rates.

Five similar modules are designed, which perform 2-D convolutions, followed by batch normalization and conclude with a ReLU activation function. The filters start with large kernel sizes in the first layer (9×9), but shrink gradually thereafter resulting in 3×3 kernel sizes at the last modules. In the first layer the filter is not up-sampled, the dilation rate equals to 1. Then dilated convolutions are performed with a progressively increased dilation rate for the rest of the modules, thus increasing the network's receptive field.

In each of the aforementioned modules, L2 regularization, even if in a small degree, is invoked for the weights as another measure against overfitting. The last layer of the network consists of a 2-D convolution with four 1×1 kernels and a softmax activation function, so that each pixel has the posterior probabilities for each class assigned (Background, Right Ventricle, Myocardium and Left Ventricle).

We also found that no concatenations, or any kind of layer aggregation, are needed to boost performance. Only few convolutional kernels and thus a small amount of trainable network parameters are able to achieve the desired outcome. The detailed architecture of our DCNN featuring 229220 parameters in total is presented in Fig. 3, while in Table I more details about the receptive field and the parameters of each layer are given. As we can notice, the kernels start densely, but expand and contract sparsely with the analogous consequences in terms of capturing context, while the receptive field gradually increases.

We consider that on the one hand the localization ability of the network is improved [14], and on the other hand the gridding artifacts are restrained. Effectively, from the second to the second-to-last layer the kernel size of the previous layer is larger than the kernel size of the current layer, and therefore the inputs in neighboring sites are not completely disjoint. We found that the network tends to learn smooth feature maps in the early layers, thus limiting artifacts due to the filter up-sampling. We obtained smoothed dilated convolutions via the kernel size, in a way that differs from that of [17].

In addition, it appears that the filters in the first layer could have interesting interpretations related to the data of the considered application domain. We show in Fig. 4 in the first row the eight more expressive learned filters of the first layer and in the second row their corresponding smooth approximations as first or second order derivatives. More precisely, among the sixteen filters, six (6) are smoothed first order derivatives, eight (8) are smoothed second order derivatives and two (2) are auto-convolutions of second order derivatives. We could say that the network learns to extract understandable and discriminative features directly relevant to the image content.

Furthermore, our experiments have shown that the network performs better and trains faster when the kernel initialization in the first layer is not left to random. Usually a typical random kernel initialization, such as the Glorot uniform [6], is employed for weight initialization, and we also do so in the other convolutional layers. However, in the first layer the kernels are initialized with the best weights that particular layer had to offer after multiple training sessions. To elaborate further, because the kernels in the first layer are quite large, we have chosen to keep their number small, so that computational cost and time are not burdened. However having a small number of kernels can not guarantee that all the desired low level features, such as edges, luminances, curves etc., are learned by the training procedure, especially when we opt for faster training times. Therefore, by observing the weights of the first layer after each training in a 5 fold cross-validation scheme and repeating this several times, we automatically selected, using a clustering algorithm,

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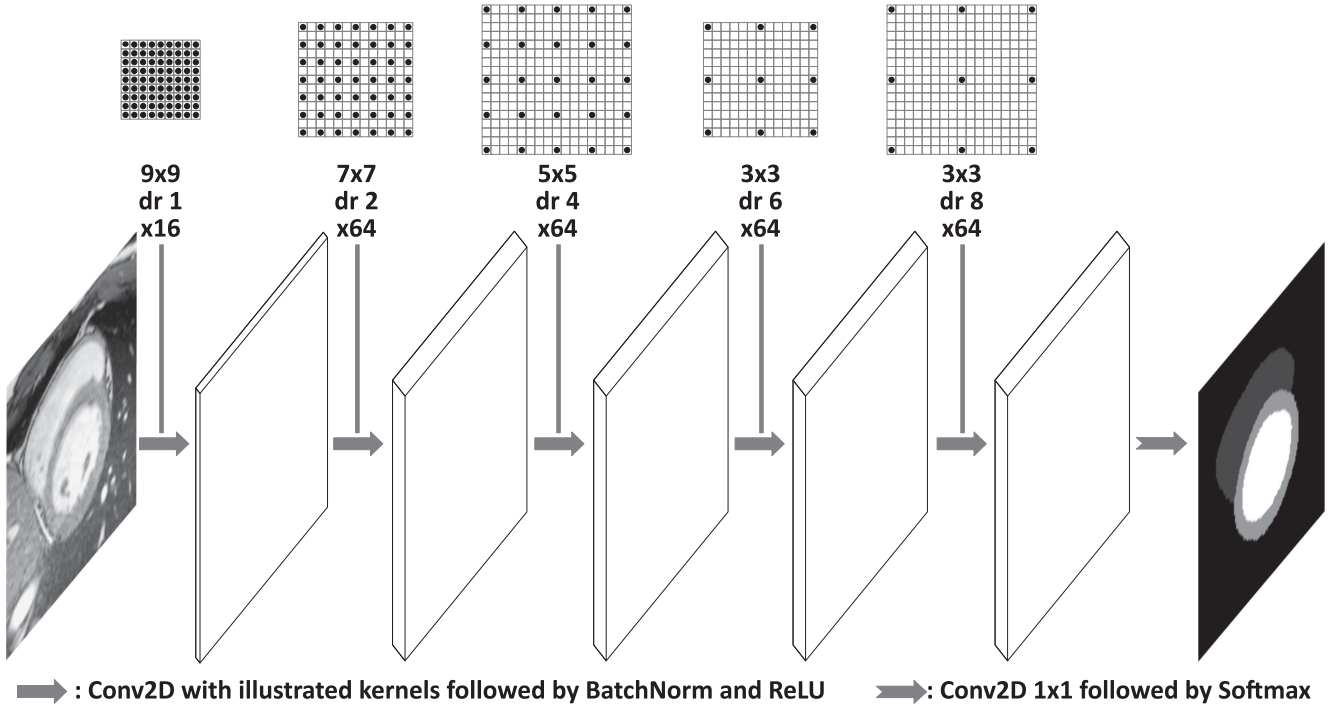


Fig. 3. Dilated Convolutional Neural Network (DCNN) architecture.

TABLE I
ARCHITECTURE DETAILS OF OUR DILATED CNN LAYERS

Convolution layer:	1	2	3	4	5	Softmax
Kernel size:	9×9	7×7	5×5	3×3	3×3	1×1
Filter nr.:	16	64	64	64	64	4
Dilation rate:	1	2	4	6	8	1
Receptive field:	9×9	21×21	37×37	49×49	65×65	65×65
Parameters:	1376	50496	102720	37184	37184	260

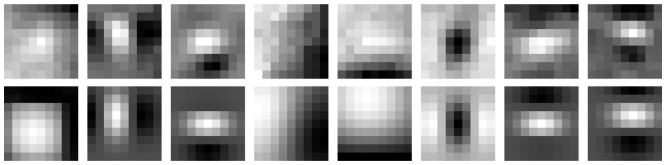
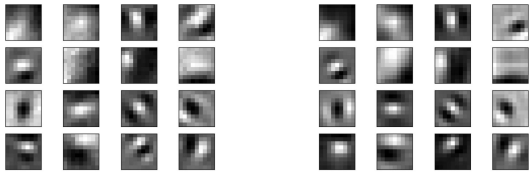


Fig. 4. Eight more expressive convolutional kernels of the first layer (1st row) and their corresponding smooth approximations (2nd row).



(a) Initial weights (b) Trained weights

Fig. 5. Sixteen 9×9 convolutional kernels of the first layer before and after training for one fold.

16 different filters, which clearly depicted desired low level features according to the training data set, as shown in Fig. 5(a). Again, these filters are not manually designed and, of course, the network then also updates these weights according to the training process to result in final filters as shown in Fig. 5(b). Since these weights are the least affected by the back-propagation procedure, the network has a consistent foundation to build on with that modification.

D. Loss Function

The network's training goal is to minimize a carefully designed loss function featuring the soft Dice loss:

$$L_D = \sum_{c=1}^{4} w(c) \left(1 - \frac{2 \sum_s Y(s,c)T(s,c) + \epsilon}{\sum_s (Y^2(s,c) + T(s,c)) + \epsilon}\right), \quad (1)$$

where the summation on s is over all the pixels on all the images in a mini-batch. Y and T denote the networks' prediction and the ground truth respectively. The class weights are set as $w = [1\ 2\ 4\ 1]$, where label '1' is for the background (BGR), '2' for the RV cavity, '3' for the Myocardium and '4' for the LV cavity. The Myocardium, and more exactly the epicardium, is the most difficult to segment, followed by the RV cavity and therefore the weights are set accordingly.

This loss is complemented by additional criteria that ensure integrity of the predictions specifically for the ACDC data set.

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The additional criteria are aiming to take in account the segments' topology, either generic for slices where both cavities are present, or when the learning of specific configurations should be enforced.

First, another term is added such that predictions for classes ($c = 2, 3, 4$) that do not exist at all in an image are penalized. If $T(u, n, c) = 0, \forall u$, where u are the image pixels of image n in the mini-batch,

$$L_C(n, c) = -\sum_u \ln\left(1 - Y(u, n, c) + \epsilon\right). \tag{2}$$

Otherwise there is no cost. A small smoothness addend ϵ is present just for proper numerical computations. To understand the importance of (2), we present its derivative used in the back-propagation process for weight updating, if $T(u, n, c) = 0, \forall u$,

$$\frac{dL_C}{dY} = \frac{1}{1 - Y(u, n, c) + \epsilon}, \tag{3}$$

that is if any of the classes RV, MYO, LV is not present, it becomes a punishing measure for emerged predictions of such classes.

Finally, a penalty for pixels belonging to the Left Ventricle that are neighbors with pixels of the background or Right Ventricle area is featured, because the LV cavity is always surrounded by the myocardium. Therefore we penalize such outcomes by the term:

$$L_V = \sum_u \sum_{t \in \mathcal{N}(s)} Y(u, n, 4)\left(Y(t, n, 1) + Y(t, n, 2)\right)), \tag{4}$$

where the first summation on u is again over all the pixels of image n in the mini-batch for the predictions of class '4' (LV) and the second summation of t is over the immediate neighborhood of pixel u of image n in the mini-batch for predictions of classes '1' and '2,' namely the background and the Right Ventricle. That way, on the one hand LV, BGR adjacency is disapproved of, and on the other, LV, RV contiguity is also discouraged. The final loss function forms then as shown in Equation (5),

$$L = L_D + \alpha L_C + \beta L_V. \tag{5}$$

The terms L_C and L_V incorporate also the same weights as described in Eq. (1). Therefore the total loss L copes with class imbalance, such that classes with heavy presence on the training set, covering larger areas, do not bias the network to false conclusions. In addition, the weights are fixed aiming to minimize the risk of erroneous classifications, taking into specific consideration the more ambiguous and in-homogeneous regions.

E. Training

Summarizing the augmentation process we start with the original ACDC training dataset consisting of 1902 cardiac MR images, pre-process them as stated in II-A and obtain another 2876 images augmented on the intensities as presented in II-B. We then randomly distribute the original pre-processed training images into 5 folds in a stratified manner, making sure each fold contains data from each one of the 5 groups (4 pathological plus 1 healthy subject groups, as stated in the ACDC challenge). For the training process each fold which constituted the training set was enriched with the corresponding intensity augmented images. The validation set remained as was, so that images in this set were never introduced to the network during the training stage, as neither did intensity augmented samples of them. Therefore we have approximately for each fold a training set consisting of 3822 images and a validation set consisting of 381 images, from a pool of a total 4778 images. Small variations over the folds occur because for each patient the number of available slices may differ.

TABLE II
TRAINING AND VALIDATION LOSS AND ACCURACY VALUES OVER ALL FOLDS, AS WELL AS THE AVERAGES

Folds:	1	2	3	4	5	Averages
Loss Tr:	0.048	0.049	0.048	0.048	0.049	**0.048**
Loss Val:	0.071	0.067	0.068	0.074	0.063	**0.069**
Acc Tr:	0.986	0.986	0.986	0.986	0.986	**0.986**
Acc Val:	0.098	0.981	0.982	0.980	0.982	**0.981**

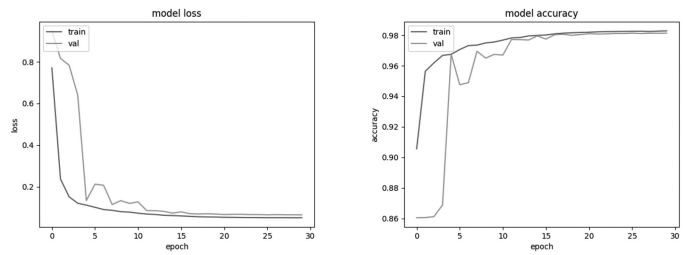

Fig. 6. Training loss and accuracy for fold 2.

For optimization purposes we chose a Stochastic Gradient Descend algorithm with momentum. We started with a learning rate of 0.04 and found out that decreasing it at each epoch by a factor of 0.8 leads to better results and faster convergence. In particular, 30 epochs were enough for convergence, where at each epoch twice the fold's training data, augmented, was fed to the network.

The mini batch size was determined to 32. Fewer images in the mini batch did not favor training and it compared in performance with bigger sized batches which in addition required more GPU memory. We randomly built the mini batch from the fold's training set, but made sure that each batch contained in any case three images which had at least one of the classes of interest (RV, MYO, LV) not present. In our experiments we noticed that the network performed poorly in cases like that and especially if none of these classes were present in the image. Such images are significantly less present in the training set and the network didn't see enough of them to learn. By this modification we present the network with more such cases resulting in improved performance. The loss and accuracy are reported in Table II, proving that the proposed augmentation scheme with the dilated CNN leads to consistent and stable performance throughout the folds. A sample of the learning curves is depicted in Fig. 6. All the folds show similar behavior during the training.

We have to emphasize that nearly 79 out of 80 images that the network trains on are augmented, since original and intensity augmented images are subject to adapted random rotations and scaling in real time during the training phase. That means that the

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network rarely has a chance to train on original, non augmented data.

Our network was able to train each fold in approximately 14 minutes on an NVidia Titan V GPU using Keras/Tensorflow. After training, inference for one image is extracted in 1 ms. In Section IV we present scores for predictions on the validation sets for each fold before the post-processing, which are already encouraging. Our post-processing method improves the segmentation results as elaborated below.

III. Prediction Post-Processing

The 5-fold cross-validated trained dilated CNN is used for computing voxel-wise probabilities for the four classes. As the training is done in 5 folds the per class probabilities are obtained as the mean values of the 5 computed from the training folds for every test image. The cardiac volume segmentation could be obtained by maximizing these probabilities voxel-wise, as they are computed on the 2-D slices. However anatomically incoherent segmentation results might occur. Our objective is to obtain a more anatomically consistent segmentation by exploiting 3-D information and known constraints concerning region properties.

At a first stage the segmentation is posed as a probabilistic optimization problem using a 3-D discrete Markov random field (*MRF*) in order to obtain a regularized label field. In this manner, we aim at capturing the local interactions between voxels and the coherence of the segmentation 3-D map.

The problem can be formulated as follows: we seek to assign a class label $l(v)$ to each node (voxel) of a graph $v \in \mathcal{V}$, so that the following cost is minimized:

$$\sum_{v \in \mathcal{V}} c_l(v) + \sum_{(v,u) \in \mathcal{E}} d(l(v), l(u)), \tag{6}$$

where $\mathcal{E}$ is the set of the graph edges. In this work for a 3-D volume a first-order model with 6 connections is employed. The graph is composed by the whole regular 3-D grid of voxels.

The singleton potentials, or priors, are based on the voxel-wise computed probabilities. Then the dissimilarities of voxels to the labels l are given by

$$c_l(v) = -\ln p_l(v). \tag{7}$$

The pairwise potentials are set according to the following weights matrix

$$D = d_0 \begin{bmatrix} 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & 9 \\ 1 & 1 & 0 & 1 \\ 5 & 9 & 1 & 0 \end{bmatrix}. \tag{8}$$

The regularization constant d_0 is data adapted, having as relevant statistics the average value on the minimum dissimilarities over the whole volume. For minimizing the MRF energy in Equation (6), we make use of the *primal-dual* method [12].

As in the ACDC data sets the identification of the most basal and apical slices is not provided, their detection is needed, because the CNN is not always able to predict them. If in the first slice (for the ED phase) or in one of the first slices (for the ES phase), possibly the most basal, no LV cavity is detected or the mean probability of the detected as LV cavity area is less than 0.7, then this slice is considered beyond the volume of interest. Such a case is shown in Fig. 7. On the left the image of a slice is shown and on the right the computed probabilities are given. In this case the mean probability of the presumed LV cavity is less than 0.7.

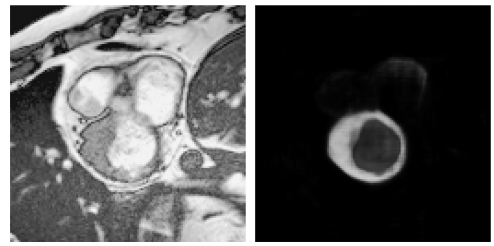

Fig. 7. Detection of false LV cavity prediction.

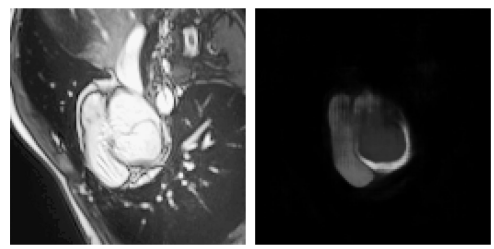

Fig. 8. Detection of false heart prediction.

If in the last slice, possibly the most apical one, there is no myocardium predicted, then this slice is also considered beyond the volume of interest.

Then, only for the diastolic phase, the largest 3-D 6-connected component is selected for the whole heart. In addition, the largest 3-D 6-connected component of the left ventricle, including the cavity and the myocardium, is selected.

For both diastolic and systolic phases a supplementary test for the most basal slice is applied based on the shape of the myocardium. A slice detected beyond the ventricular volume by this test is shown in Fig. 8. Again in this Figure the image and the corresponding computed probabilities are illustrated.

The final segmentation map for the volume is obtained at the end of the post-processing steps.

IV. Results and Discussion

In this section we present and discuss results from our experiments. As stated previously, we used a 5 fold cross-validation scheme for our training. It allowed us to observe the network's consistency in performance.

At first, we measured the Dice score directly on the predictions for the validation sets using the trained weights for each of the corresponding folds. That way we could see how each component, added or altered, affected the network's training and if it contributed to a better end result or not. Then, the Dice scores were measured for each patient for the LV, RV and MYO classes for each one of the cardiac phases, end diastole

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TABLE III
DICE SCORES FOR LV, RV AND MYO ON THE VALIDATION SETS

Method	Mean Dice LV ED	Mean Dice RV ED	Mean Dice MYO ED	Mean Dice LV ES	Mean Dice RV ES	Mean Dice MYO ES
*DCNN	0.822	0.761	0.510	0.719	0.652	0.601
DaDCNN-00	0.954	0.920	0.869	0.898	0.824	0.878
DaDCNN-01	0.955	0.916	0.868	0.902	0.845	0.880
DaDCNN-10	0.958	**0.924**	0.872	0.912	0.846	**0.885**
DaDCNN-11	**0.959**	**0.924**	**0.873**	**0.918**	**0.849**	0.884
*Unet-2	0.822	0.779	0.513	0.728	0.677	0.605
DaUnet-2	0.963	0.935	0.888	0.921	0.856	0.898
*Unet-1	0.825	0.778	0.509	0.732	0.684	0.605
DaUnet-1	0.959	0.921	0.877	0.915	0.839	0.890
*DeepLabv3+	0.914	0.824	0.746	0.825	0.686	0.771

(ED) and end systole (ES). In Table III we present these metrics for the proposed CNN using only classical augmentation techniques (random rotations $[0, \pi]$, random scaling [0.7, 1.3] and minor elastic deformations) and the plain soft Dice loss as in Eq. (1). Then we show how much improvement our adapted augmentation together with our designed loss function given in Eq. (5) achieved (the Data-adapted Dilated CNN-DaDCNN). It is shown gradually from rows 3 to 5 that the Data-adapted augmentation, as well as each component in the specially designed loss function improve performance. In DaDCNN-00 only the Dice loss function is used, in DaDCNN-01 the loss function is $L_D + \beta L_V$, in DaDCNN-10 it is $L_D + \alpha L_C$ and in DaDCNN-11 the whole loss function of Eq. (5) is employed.

Additionally, we report the same scores obtained from predictions resulted by a U-net, trained without and with the proposed adapted augmentation and final loss function. Two versions of the U-net have been implemented and tested, one with twice as many parameters as the DCNN (Unet-2) and one with almost the same numbers of parameters (Unet-1). Again, improvement is achieved by utilizing the adapted augmentation and designed loss function. Our implementation of the U-net consisted of three max pooling stages, reducing the spatial dimensions from 120×120 to 15×15, and respective transposed convolution steps to return to original dimensions. Each layer consists of two successive convolutions, batch normalization and ReLU activation and, skip connections are also applied to transfer context from the the down-sampling path to the up-sampling path. To achieve a number of parameters comparable to the DCNN we used a second scheme and reduced the number of filters. The network was trained using an Adam optimizer with decay, for 30 epochs as well, batch size 32 and learning rate 0.003. Each fold trained in 15 min, respectively 7.5 min, while inference time for one image was 1 ms in both schemes.

Finally, the DeepLabv3+ model [5] was also considered, being the state of the art in image segmentation nowadays. We used the ResNet-v1-50 as a backbone and trained the model from scratch using our pre-processed ACDC data set for 300000 iterations. No further modifications were done. The DeepLabv3+ managed to outperform by far all approaches with the standard soft Dice loss function and without the utilization of the adapted augmentation (rows marked with a star in Table III).

Taking into account the Data-adapted augmentation (rows starting with Da in Table III) we can see that the DCNN compares and even outperforms in some metrics the U-net, using our data augmentation method, with more or less the same amount of parameters.

In Tables IV, V and VI the evaluation of the proposed method on the ACDC test data set is given, as submitted to the Post-2017-MICCAI-challenge testing phase.[2] The metrics are those used for evaluation in the ACDC challenge. There are four distance measures, two for each phase, and six clinical indice metrics for the LV cavity, the RV cavity and the myocardium. In all the Tables the bias, the standard deviation (std) and the mean absolute error concerning volumes are given in mL/m^2, the myocardium mass is given in g/m^2 and the Hausdorff distance is given in mm.

A comparison is given with the best algorithms submitted to the challenge as they appear in the leader-board of the challenge[3] [10], [11], [13], [21]. We present briefly the main architectural feature of these four algorithms. Isensee *et al.* [10] used a network architecture inspired by the 3-D U-net. Khened *et al.* [11] used a densely connected fully convolutional network instead of the skip connections used in the U-net architecture. Zotti *et al.* [21] used a multiresolution Gridnet architecture, which could be considered as an extension of the U-net architecture. Painchaud *et al.* [13] proposed an adversarial variational autoencoder for guaranteeing anatomically plausible segmentations.

In the case of the LV cavity we have obtained the best score on the correlation coefficient and the standard deviation for the ejection fraction and the volume in the ED phase. In addition we have obtained the best score on the Dice coefficient for the LV cavity in the ED phase. We have also obtained the best result on the correlation coefficient and the standard deviation for myocardium mass. For the RV cavity and the myocardium ES phase we have obtained less good results, but comparable to those of other well performing deep learning methods.

We compare also the proposed method with our implementation of a U-net and the methods of [10], [11] on the reliable clinical metric of the mean absolute error for the LV/RV ED volume, the LV/RV Ejection Fraction, and the myocardium mass. The comparison is given in Table VII, where the metrics for [10], [11] are as published in [1]. For our algorithm we have measured the performance on mean absolute error for the prediction resulting from the proposed neural network without post processing, the result with only the 3-D MRF model, with only the other post-processing modules, and the whole algorithm described in Section III. We can see that the post-processing improves the mean absolute error for the five criteria. Concerning the two terms of the post-processing, we found that the more costly 3-D MRF has limited impact, only on the RV Volume in ED phase. The specific post-processing modules addressing anatomical inconsistencies seem to have a clearly

[2][Online]. Available: https://acdc.creatis.insa-lyon.fr/#submission/5ecb92156a3c7704987f95c7

[3][Online]. Available: https://acdc.creatis.insa-lyon.fr/description/results.html

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TABLE IV
METRICS ON TEST DATA SET FOR THE LEFT VENTRICLE

Method	Mean Dice ED	Mean Dice ES	Mean Hausdorff ED	Mean Hausdorff ES	EF corr coef	EF bias	EF std	Volume corr coef ED	Volume bias ED	Volume std ED
Isensee	0.965	**0.933**	**5.6**	**6.3**	0.992	**0.3**	2.8	0.997	1.6	5.8
Zotti	0.964	0.912	6.2	8.4	0.990	-0.5	3.1	0.997	3.7	5.1
Painchaud	0.961	0.911	6.1	8.3	0.990	-0.5	3.2	0.997	3.8	5.2
Khened	0.964	0.917	8.1	9.0	0.989	-0.5	3.4	0.997	**0.6**	5.5
DaDCNN	**0.967**	0.928	6.4	7.6	**0.993**	-0.4	**2.7**	**0.998**	2.0	**4.6**

TABLE V
METRICS ON TEST DATASET FOR THE RIGHT VENTRICLE

Method	Mean Dice ED	Mean Dice ES	Mean Hausdorff ED	Mean Hausdorff ES	EF corr coef	EF bias	EF std	Volume corr coef ED	Volume bias ED	Volume std ED
Isensee	**0.946**	**0.904**	**8.8**	**11.4**	**0.925**	-3.0	**5.1**	**0.991**	2.1	**9.0**
Zotti	0.934	0.885	11.0	12.6	0.869	**-0.9**	6.8	0.986	2.4	11.5
Painchaud	0.933	0.884	13.7	13.3	0.865	**-0.9**	6.9	0.986	**2.1**	11.4
Khened	0.935	0.879	14.0	13.9	0.858	-2.2	6.9	0.982	-2.9	12.6
DaDCNN	0.936	0.889	13.3	14.4	0.894	-1.3	6.1	0.990	**0.9**	9.7

TABLE VI
METRICS ON TEST DATA SET FOR THE MYOCARDIUM

Method	Mean Dice ED	Mean Dice ES	Mean Hausdorff ED	Mean Hausdorff ES	Vol ES corr coef	Vol bias ES	Vol std ES	Mass ED corr coef	Mass bias ED	Mass std ED
Isensee	**0.896**	**0.919**	**7.6**	**7.1**	**0.989**	-3.4	**7.9**	0.986	-4.0	8.6
Zotti	0.886	0.902	9.6	9.3	0.980	1.2	10.9	0.986	**-1.8**	8.6
Painchaud	0.881	0.897	8.6	9.6	0.979	**0.3**	11.1	0.987	-2.9	8.4
Khened	0.889	0.898	9.8	12.6	0.979	-2.6	11.0	0.990	-2.9	7.5
DaDCNN	0.891	0.904	8.3	9.6	0.983	-2.1	10.1	**0.992**	-2.9	**6.5**

TABLE VII
MEAN ABSOLUTE ERROR ON THE TEST DATASET

Method	LV EDV	LV EF	RV EDV	RV EF	Myo mass
DaDCNN+MRF+post	3.4	**1.9**	**7**	**4.3**	**5**
DaDCNN+MRF	4.8	3.4	7.2	5.4	6.1
DaDCNN+post	**3.3**	**1.9**	7.4	4.4	5.1
DaDCNN	4.7	3.4	7.4	5.4	6.2
DaUnet+MRF+post	**3.3**	2.1	7.7	4.6	6.1
DaUnet	3.9	2.3	8.1	4.8	6.2
Isensee	5.1	2.1	7.9	4.7	7.3
Khened	4.2	2.5	8.4	5.3	6.3

measurable impact. We have also measured the performance of our implementation of the U-net, with and without MRF and post-processing, and found that some improvement has been also obtained. Finally, our method gives the best result, among the up to now known results on this data set.

V. CONCLUSION

In this work our main objective was to prove that a simple CNN could achieve top performance if its design and training is domain-driven.

For this purpose, a cascaded dilated convolutional neural network is designed for semantic segmentation of MR cardiac images. The first layer starts with larger convolutional kernels and no dilation, but kernel sizes become smaller in opposite to the dilation rate which increases, as the architecture deepens. The network has fewer parameters compared to other deep learning architectures designed for that task, and needs a small number of epochs to train. Even if the proposed network has a small number of features and parameters, we think that it would be possible to further simplify the network, probably by accordingly designed training. This task could be facilitated by the ascertainment that the first layer filters of the proposed network have an interesting interpretation.

The training process is assisted by carefully adapted data augmentation that relies on simple image transformation techniques such as rotation and scaling. Prior to training, a specially designed intensity augmentation takes place to enrich the training data set. Based on the distribution of the probability density functions of the Left Ventricle and the myocardium, the contrast reduction is adjusted to produce more samples. Our results show that this is the decisive key for the top performance obtained in all metrics.

Finally, since the network does not take into account volume information, but produces predictions for each image independently, a post-processing step incorporates anatomical constraints to produce a more consistent segmentation map for the volume.

Even though we have achieved top performance in many evaluation criteria, there is still room for improvement, especially with regard to the accuracy of the RV localization and the myocardium ES phase. We plan to work on adapted data

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augmentation for the RV as well, as up until now only the LV was specifically used for that. We would like to pursue the extended testing of our adapted augmentation with other network architectures, as we did with a U-net implementation of our own, to establish its contribution in improving performance. Furthermore, since classic augmentation techniques did not improve ES results, we have begun to investigate the use of Generative Adversarial Networks (*GANs*) introduced by [7] to produce augmented samples which are not manually designed but learned by a neural network. A task related Variational Autoencoder is also implemented in [13] making use of adversarials.

At last, we plan to test our approach on other data sets to determine if the adaptation could be applied in a wide range of similar data from different clinical centers and different scanners. We also intend to develop deep learning techniques to substitute pre- and post-processing. An NN assembly or a single architecture trained end-to-end without any human intervention could undertake the problem for automated, fast and accurate semantic segmentation on cardiac MR volumes.

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